

XQAS: eXplainable Quantum Architecture Search

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Abstract. Automating the design of quantum circuit design via Quantum Architecture Search (QAS) can potentially harness the utility of near-term quantum devices. Reinforcement Learning (RL) based strategies offer a useful strategy for QAS. However, it still has to deal with two persistent bottlenecks: parameter inefficiency of conventional agents alongside the black-box nature of the solutions they obtain. In this work, we aim to overcome these bottlenecks by introducing a framework that combines two recently developed, complementary approaches for addressing these issues. First, we replace the Multi-Layer Perceptron with the efficient Kolmogorov-Arnold Network (KAN) to improve the agent’s performance. The KAN-based agent could provide a significantly higher success rate, greater noise resilience, along with compact circuits for a range of state preparation and quantum chemistry tasks. Following this, we analyze the admissible circuits produced by the agent by borrowing tools from Quantum Information Theory. Our analysis provides insightful design principles, including an entanglement-driven phase transition that occurs during state diagonalization. The combined approach presented in this work integrates an improved RL-agent along with physical analysis, thereby advancing automated quantum circuit design.

Keywords: Quantum Architecture Search · Kolmogorov-Arnold Network · Reinforcement Learning · Variational Quantum Algorithms · Quantum Information Theory

1 Introduction

The application of machine learning approaches to automate the design of quantum algorithms [Sar24] can potentially harness the utility of near-term hardware [BCLK⁺22]. One such approach termed Quantum Architecture Search (QAS) implements an optimization routine to discover effective quantum circuits for tasks of interest. A recent QAS routine called RL-QAS employs a Reinforcement Learning (RL) agent to navigate the circuit configuration space [OTM⁺21, FNML21]. A promising use-case of RL-QAS is to find optimal parameterized quantum circuits for Variational Quantum Algorithms [CAB⁺21] (VQAs). However, it is limited by the efficiency of the RL agent and the interpretability of the solutions obtained by the agent.

The first bottleneck arises from the efficiency of the RL-QAS agent being dependent on the internal neural network (typically a Multi-Layer Perceptron, i.e., MLP) acting as a policy-maker. In many cases, the quality and efficiency of the solution may suffer as the underlying MLPs may struggle with the complex optimization landscapes of the quantum problem. In a recent work, the authors of [LWV⁺24] have introduced Kolmogorov-Arnold Networks (KANs) as an alternative

to MLPs in classical machine learning. Inspired by the Kolmogorov-Arnold representation theorem, KANs utilize learnable spline-based activation functions on the network edges. Consequently, these networks [IHK⁺24, JHCG25, WML⁺25, CMC⁺25] demonstrate greater accuracy and expressivity while requiring smaller network sizes.

The second bottleneck arises from the interpretation of the effective solution circuits generated by the RL-agent. This bottleneck arises as we lack insight into why RL-QAS circuits are effective. This lack of understanding of strategies learned by the agent limits the generalizability of the solutions obtained. A thorough analysis of the solution circuits from the perspective of Quantum Information Theory (QIT) is therefore needed to understand how the RL-agent operates.

Contribution In this work, we address these bottlenecks by introducing XQAS, a unified framework integrating the QAS approach with a rigorous QIT methodology. Our contribution can be summarized as follows:

1. We tackle the efficiency bottleneck by replacing the agent of the Double Deep Q-Network (DDQN) with a KAN, thereby introducing KANQAS. We benchmark the KANQAS architecture on quantum state preparation tasks (Bell and GHZ states) and quantum chemistry problems (ground-state energy of H_2 and LiH). We observe that the KAN-based agent (a) achieves higher success rates, (b) is more robust to noise, and (c) discovers more compact circuits with substantially fewer network parameters.
2. We tackle the interpretability problem by analysing Quantum Information Theoretic metrics of the solutions obtained from the RL-QAS framework [KBO⁺24] when solving the Variational Quantum State Diagonalization (VQSD) problem [LTOJ⁺19]. We observe that (a) the entanglement-driven phase transition dictates the agent’s design strategy, and (b) correlations in conditional quantum entropy explain the algorithm’s performance limitations in resolving a state’s full eigenspectrum.

The approach presented in this work presents a paradigm for combining the black-box optimization routine of QAS with physical understanding, thereby potentially moving the field towards a future where automated discovery yields solutions along with scientific insight.

2 Background and preliminaries

2.1 Reinforcement learning for quantum architecture search

We formulate the Quantum Architecture Search (QAS) problem as a Markov Decision Process (MDP) within the framework of Reinforcement Learning (RL) [SB18]. We define the MDP as a tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$ where:

- (a) **State ($\mathcal{S}$):** the quantum circuit which starts as an identity gate on all the qubits. The agent appends gates to it following a policy.
- (b) **Action space ($\mathcal{A}$):** it is quantum gates that the RL agent adds to the circuit. In this work, we consider the RL agent that uses single-qubit rotations, CNOTs, and single-qubit Pauli gates.
- (c) **Transition (P):** is basically the state before the application of an action (i.e., a gate) and after an action.
- (d) **Reward (R):** a value calculated based on the cost function of the Variational Quantum Algorithms. **Discount Factor (γ):** a factor $\gamma \in [0, 1]$ that balances the trade-off between the immediate and future rewards obtained by the RL-agent.

The RL agent’s objective is to learn a policy $\pi(a|s)$ which maximizes the expected cumulative discounted reward. The choice of RL agent is completely problem specific [IDTK25]. In this work, the agent we consider is a Double Deep Q-Network (DDQN) [OTM⁺21]. DDQN is a value-based method that learns an action-value function $Q(s, a)$ to estimate the importance of action a on the state s at step t . Such a structure consists of two neural networks: (a) primary, and (b) target network for enhancing the stability of the learning algorithm.

2.2 Kolmogorov-Arnold networks (KANs)

Recent trends in reinforcement learning (RL)-based quantum architecture search (QAS) [OTM⁺21] utilize multi-layer perceptrons (MLPs). These networks are grounded in the universal approximation theorem [HSW89], composed of linear transformations followed by fixed, nonlinear activation functions. Learning proceeds via adjustments to the weights of the linear layers. However, the selection of appropriate activation functions and the proliferation of trainable activation parameters remain significant challenges, especially when interpretability of these networks is desired.

To address the challenge of parameter scalability and interpretability in MLPs, Kolmogorov-Arnold Networks (KANs) [LWV⁺24] have been recently introduced. KANs are based on the Kolmogorov-Arnold representation theorem (KART) [Kol57], which asserts that any real-valued, smooth, and continuous multivariate function $g(\mathbf{t}) : [0, 1]^n \rightarrow \mathbb{R}$ can be expressed as a finite superposition of univariate functions [Kol57]:

$$g(\mathbf{t}) = \sum_{j=1}^{2n+1} \Psi_j \left(\sum_{k=1}^n \psi_{jk}(t_k) \right), \quad (1)$$

where $\Psi_j : \mathbb{R} \rightarrow \mathbb{R}$ and $\psi_{jk} : [0, 1] \rightarrow \mathbb{R}$. This result indicates that any continuous multivariate function on a bounded domain may be written as a composition of continuous single-variable functions, reducing the challenge of learning high-dimensional mappings to the problem of optimizing a polynomial number of univariate functions.

KANs are organized in layers, where each layer is defined by a matrix Ψ whose elements are trainable univariate spline functions $\{\psi_{jk}(\cdot)\}$, with $j = 1, \dots, n_i$ and $k = 1, \dots, n_o$. Here, n_i and n_o denote the input and output dimensions for that layer [LWV⁺24]. In the Kolmogorov-Arnold representation theorem (KART), the construction can be understood as a two-layer KAN: an inner layer where $n_i = n$ and $n_o = 2n + 1$, and an outer layer where $n_i = 2n + 1$ and $n_o = n$. The overall shape of a KAN is specified by listing the number of units in each layer, $[n_1, \dots, n_k]$, with k indicating the number of layers. For KART, this structure is given as $[n, 2n + 1, 1]$. A deeper KAN simply uses more layers in sequence:

$$\mathbf{z} = \text{KAN}(\mathbf{t}) = (\Psi_k \circ \Psi_{k-1} \circ \dots \circ \Psi_1)\mathbf{t}. \quad (2)$$

Crucially, because all component functions are differentiable, KANs are compatible with standard backpropagation methods for training [RDGC13]. For clarity, we describe a depth-2 KAN architecture as $[n_i, n, n_o]$, excluding the input nodes from the depth count. The input and output layers contain n_i and n_o nodes, respectively, corresponding to overall input and output dimensions, with n representing the number of hidden units.

Unlike traditional neural networks, KANs place learnable, nonlinear activation functions on the edges instead of nodes. These edge functions are parameterized as B-splines, allowing the network to flexibly adapt its activations. This approach enables KANs to efficiently capture complex, compositional relationships in data, often with significantly fewer parameters than conventional MLPs.

2.3 Quantum information-theoretic metrics

To analyze the quantum circuits generated by the learning agent for solving a particular task of interest, we use the following tools from Quantum Information Theory:

(a) **Fidelity** [Wil13]: the fidelity (or overlap) between a pure state of interest $|\psi\rangle$ and the generated state ρ_{gen} is given by $F = \langle\psi|\rho_{\text{gen}}|\psi\rangle$. For quantum state preparation tasks, fidelity quantifies the probability of successfully preparing a state.

(b) **Purity** [Wil13]: the purity of a quantum state ρ is given by $\text{Tr}(\rho^2)$. Pure states have purity $\text{Tr}(\rho^2) = 1$ while mixed states have purity $\text{Tr}(\rho^2) < 1$.

(c) **Concurrence**: for a bipartite state ρ_{AB} , the amount of entanglement can be quantified via concurrence [HW97, Woo98] defined as:

$$C^{\rho_{AB}} = \max\{0, \lambda_1^e - \lambda_2^e - \lambda_3^e - \lambda_4^e\} \quad (3)$$

where λ_s^e are the square root of the eigenvalues of $\rho_{AB}\tilde{\rho}_{AB}$, arranged in descending order. The term $\tilde{\rho}_{AB} = (\sigma_y \otimes \sigma_y)\rho_{AB}^*(\sigma_y \otimes \sigma_y)$ is the spin-flipped state of ρ_{AB} and σ_y is a Pauli spin matrix. For the pure state $\rho_{AB} = |\psi\rangle\langle\psi|_{AB}$, the concurrence can be simplified as

$$C^{|\psi\rangle_{AB}} = \sqrt{2[1 - \text{Tr} \rho_A^2]}, \quad (4)$$

with $\rho_A = \text{Tr}_B \rho_{AB}$ denoting a subsystem of the two-qubit joint state.

(d) **Conditional quantum entropy** [Wil13] The conditional quantum entropy $S_{A|B}^{\rho'_{AB}}$ of the state ρ'_{AB} is the difference between the joint quantum entropy $S_{AB}^{\rho'_{AB}}$ and the marginal entropy $S_B^{\rho'_{AB}}$ given by:

$$S_{A|B}^{\rho'_{AB}} := S_{AB}^{\rho'_{AB}} - S_B^{\rho'_{AB}}. \quad (5)$$

Operationally, the conditional entropy of the state ρ'_{AB} signifies the mixedness of the marginal state ρ'_A (ρ'_B) compared to ρ'_{AB} .

3 XQAS framework

In this work, we consider a DDQN agent-based framework tasked with constructing a quantum circuit of interest. The choice of DDQN is logical as it is shown [IDTK25] to provide the most compact circuit in variational quantum optimization. In this section, we highlight some of the specifics of implementation for the agent's architecture and the subsequent analysis protocol.

3.1 KANQAS architecture

The core component of the KANQAS agent is its deep Q-network, which we implement as a KAN (in one part of the work) in place of an MLP. We now present some specifics of the implementation.

(a) **State representation**: We encode the quantum circuit as a three-dimensional tensor of dimensions $D_{\text{max}} \times N \times (N + N_{1q})$. The dimensions correspond respectively to the maximum circuit depth, number of qubits, and the total number of possible gate operations. Note that a value of 1 at tensor position (d, q, g) indicates a gate g is applied at depth d to qubit(s) q . The one-hot encoding we use in this work provides a unique and unambiguous state representation for the network.

(b) **Reward formulation for state preparation:** An objective is to prepare a target quantum state with high fidelity. The reward function is structured to provide a dense signal for progress while strongly incentivizing task completion.

$$R = \begin{cases} \mathcal{R}, & \text{if } F(s_t) \geq 0.98 \\ F(s_t), & \text{otherwise} \end{cases}$$

In the above equation, the hyperparameter $\mathcal{R}$ is set to a value significantly greater than 1 to create a strong bias for the agent to surpass the target fidelity threshold, which we set as 0.98.

(c) **Reward formulation for quantum chemistry:** Another objective is to determine the energy corresponding to the ground state energy of a molecule via the Variational Quantum Eigensolver [CAB⁺21]. Such a task corresponds to minimizing the cost function $C_t = \langle H \rangle$. We structure the reward as

$$R = \begin{cases} 5 & \text{if } C_t < \xi, \\ -5 & \text{if } t \geq D_{\max} \text{ and } C_t \geq \xi, \\ \max\left(\frac{C_{t-1}-C_t}{C_{t-1}-C_{\min}}, -1\right) & \text{otherwise} \end{cases}$$

The piecewise reward function provides a large terminal reward for achieving chemical accuracy ξ along with providing a large penalty for failure. To enable progress towards energy reduction, at the intermediate steps, the reward function provides a step-wise reward proportional to the progress in energy reduction.

3.2 RL-VQSD framework

Let us consider applying RL-QAS methodology to the Variational Quantum State Diagonalization (VQSD) problem, where the objective is to find a circuit that transforms a given state into its eigenbasis.

(a) **Task and cost function:** The agent’s task is to find an unitary transformation $U(\theta)$ represented as a quantum circuit, which diagonalises an initial state ρ . The agent’s training is guided by purity-based cost function given by

$$C_t(\theta) = \text{Tr}(\rho^2) - \text{Tr}((U(\theta)\rho U^\dagger(\theta))^2). \quad (6)$$

The learning agent aims to minimize the cost $C_t(\theta)$.

(b) **Post-hoc analysis:** After the agent is trained and has successfully generated a set of circuits, we perform a detailed analysis borrowing tools from Quantum Information Theory. For each of the generated circuit, we observe the concurrence and conditional entropy of the quantum state at various stages of its evolution. By analyzing the distributions and correlations of these metrics across an ensemble of circuits, we aim to extract general principles and patterns that characterize the successful architectures that were obtained by the agent.

4 Experimental setup

For both MLP and KAN configurations, The DDQN agent was trained using standard hyperparameters. For more details of the hyperparameters used in the training, please refer to [KSS24] and [SSK24]. We next enumerate the tasks that we consider in this work.

Problem	Network	Configuration	Spline and grid	Parameter count
Bell state prep. (noiseless)	KAN	$\text{KAN}_{2,2}$	B-spline, $k = 3, G = 5$	1728
	MLP	$\text{MLP}_{4,30}$	NA	4782
GHZ state prep. (noiseless)	KAN	$\text{KAN}_{2,3}$	B-spline, $k = 3, G = 5$	5562
	MLP	$\text{MLP}_{4,30}$	NA	11181
Bell state prep. (noisy)	KAN	$\text{KAN}_{3,10}$	B-spline, $k = 4, G = 5$	9540
	MLP	$\text{MLP}_{4,100}$	NA	29912
H_2 ground state	KAN	$\text{KAN}_{4,3}$	B-spline, $k = 4, G = 5$	5.525×10^4
	MLP	$\text{MLP}_{6,1000}$	NA	5.150×10^6
LiH ground state	KAN	$\text{KAN}_{4,3}$	B-spline, $k = 4, G = 5$	5.525×10^4
	MLP	$\text{MLP}_{4,100}$	NA	1.348×10^5

Table 1. KAN requires $2 - 3\times$ fewer parameters as compared to MLP for quantum state preparation tasks. For solving the ground state of LiH and H_2 molecules, KAN utilised $2.4 - 93\times$ fewer parameters as compared to MLPs. KAN is also able to solve these problems with more compact parameterized quantum circuits as compared to MLP. We denote the networks by their depth (L) and the number of neurons (N). For an instance, the network $\text{KAN}_{2,3}$ has depth of 2 and has 3 hidden neurons in the hidden layer. The same notation is also applied to $\text{MLP}_{L,N}$ networks.

(a) **Benchmarking:** We evaluated the proposed framework on three benchmark tasks: (1) preparation of 2-qubit Bell and 3-qubit GHZ states, (2) ground-state energy estimation for H_2 and LiH molecules via Variational Quantum Eigensolver, and (3) Variational Quantum State Diagonalisation for a range of 2-qubit states with varying initial entanglement.

(b) **Noise model:** To assess robustness, we introduced a noise model that combines bit-flip errors and depolarizing channels. The noise parameters were set to $p_x = 0.3$ for bit-flips and $p_{\text{dep}} = 0.2$ for the depolarizing channel, representing a significantly noisy environment.

(c) **Baselines:** The performance of the KANQAS agent was benchmarked against identically configured DDQN agents using standard MLP architectures. Such MLP architectures had ReLU, LeakyReLU and ELU activation functions [SSA17].

5 Results and analysis

5.1 Performance and efficiency gains with KANQAS

We next present results from numerical experiments demonstrating the advantage of replacing MLPs with KANs in RL-QAS routines.

Improved Success Rate and Noise Robustness We present in Tab. 1 the details of KAN and MLP configurations used when dealing with problems in quantum state preparation and quantum chemistry.

For noiseless state preparation tasks, we observe that the KAN-based agent achieves success rates of a factor of 2 to 5 greater than that obtained by MLP-based agents. This advantage is particularly visible when preparing the 3-qubit GHZ state. The KAN-based agent could obtain a success probability of 48.41% in its final training phase. In comparison, the MLP (ReLU)-based

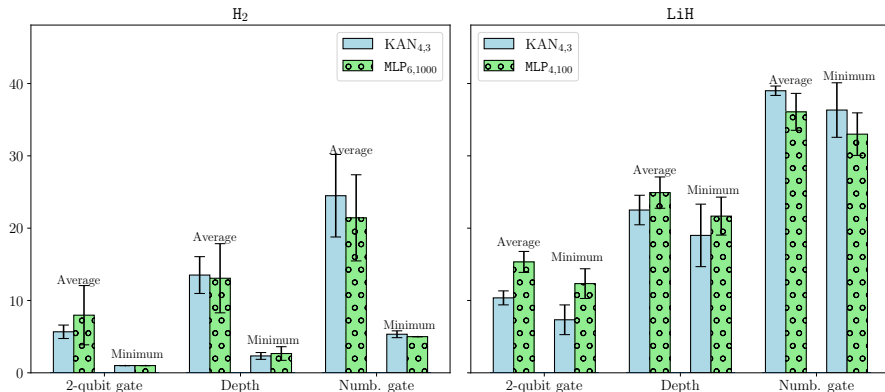


Fig. 1. Comparison of KAN and MLP to optimize parameterized quantum circuits (PQC) for 4-qubit LiH and H₂ molecules. The plots present the average and best-performing seeds over different initializations, which achieve the desired chemical accuracy of 1.6×10^{-3} Hartree. We observe that KAN_{4,3} achieves the desired accuracy using 5.52×10^4 trainable parameters thereby demonstrating superior parameter efficiency as compared to MLP_{6,1000} which requires 5.15×10^6 parameters.

and MLP (LeakyReLU)-based agents could obtain success probability of 38.23% and 38.61% respectively. These observations highlight the effectiveness of KAN at navigating the complex search space for identifying optimal parametrised quantum circuits for tasks of interest.

Next, considering noisy simulations with bit-flip and depolarizing errors, the KAN-based agent achieved a final state with a fidelity of 73.28%. In comparison, the MLP (LeakyReLU) agent obtained a comparable final state with a fidelity of 50.05% while another MLP (ReLU) agent obtained a fidelity of 50.05%, which is comparable to random guessing. We thus note that adaptive, spline-based activation functions in KAN may be robust against noise-based stochasticity.

Parameter and circuit complexity reduction Considering the task of estimating the ground state of molecular systems via VQE, we observe in Fig. 1 that the KAN-based agent required a network with 5.150×10^6 parameters to obtain quantum circuits achieving chemical accuracy for the H₂ molecule. In contrast, the MLP-based agent required a network of 5.150×10^6 parameters to reach the same performance level. This reduction of nearly two orders of magnitude in model size demonstrates the efficiency of KANs in this task. Furthermore, the circuits discovered by the KAN agent utilized fewer 2-qubit gates, providing a critical advantage for execution on NISQ hardware.

5.2 Physical insights from RL-VQSD

We next analyse the quantum circuits generated by the MLP based RL-QAS agent for solving the variational quantum state diagonalisation task.

Entanglement phase transition Consider the change in concurrence when evolving through the admissible ansatz obtained by the RL-agent for this task. We observe in Fig 2 that the entanglement properties of the quantum state influences the effectiveness of ansatz structures. Low values

of entanglement (below 0.322) allows for exploiting the variability of entanglement bounds. However, wide entanglement bounds provide ansatz flexibility. Such bounds can generate both highly entangled or nearly separable states starting from $|0\rangle^{\otimes 2}$.

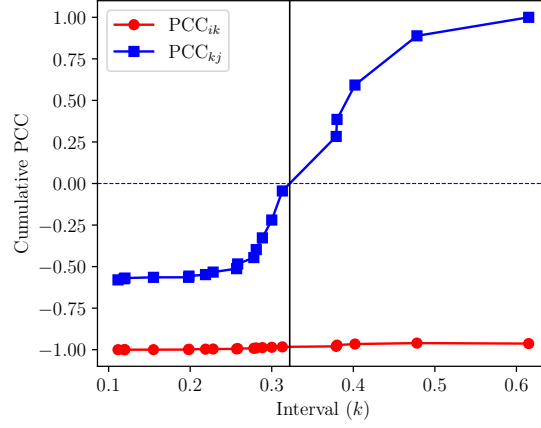


Fig. 2. We present the phase transition of concurrence bound correlation for the admissible RL-VQSD ansatz. We split the input state concurrence into intervals $u_1 := [i, k)$ and $u_2 := [k, j]$ with k as a free parameter. We plot the Pearson correlation coefficients PCC_{ik} and PCC_{kj} for the respective intervals. The phase transition point is observed at $k = 0.322$. We note at the phase transition point PCC_{kj} changes from mild anti-correlation to mild correlation while PCC_{ik} remains strongly anti-correlated.

(a) **Low-entanglement regime:** In this regime, the bounds on the concurrence of output state were found to be strongly anti-correlated, with a Pearson Correlation Coefficient (PCC) value of approximately -0.982 . This implies the following structural trade-off: circuits effective at preserving high entanglement (useful for finding one subset of eigenvalues) are fundamentally unsuited for collapsing it, and vice versa.

(b) **High-entanglement regime:** Above this threshold, the strict trade-off disappears. The correlation becomes weakly positive ($PCC \approx 0.049$), indicating that for highly entangled states, the circuit structures required to find different eigenvalues are no longer mutually exclusive.

We observe in Fig. 2 a phase transition at initial state concurrence of 0.322, which governs the bounds on concurrence of the final state. This phase transition is also reflected in circuit complexity. Ansatzes designed for states in the high-entanglement regime are more NISQ-friendly, with an average depth of 13.23 and 18.5 gates, compared to an average depth of 13.9 and 19 gates for the low-entanglement regime.

Conditional entropy We observe a decrease in the correlation between the conditional entropy of the first and second qubits with an increase in the magnitude of the two largest eigenvalues. Conversely, the correlation increases with the increase in magnitude of the two smallest eigenvalues. A single, fixed-structure ansatz prepared by the agent cannot simultaneously satisfy these two

opposing requirements. This inherent structural conflict within the problem space explains the observed performance limitations of the Variational Quantum State Diagonalisation algorithm.

6 Discussion

We present and analyse a unified framework for Quantum Architecture Search combining the architectural efficiency of KANs with the analytical depth of Quantum Information Theory. Our findings demonstrate that the KANQAS approach yields RL agents that are more parameter-efficient, robust to noise, and effective at discovering high-quality quantum circuits for key quantum-computational tasks. The RL-QAS framework also provides a methodology for interpreting the solutions proposed by the RL-agent. Specifically, it reveals fundamental physical principles such as entanglement-driven phase transitions alongside the role of inter-qubit correlations in algorithmic performance. Our approach to QAS is thus efficient, transparent, and physically grounded. By bridging the gap between automated search and physical understanding, this unified framework represents a significant step toward addressing the critical challenge of quantum circuit design in the NISQ era and beyond.

Another promising future direction would be to integrate the quantum circuit cutting methodology with the architectural efficiency of KANs, along with the analytical depth of quantum information theory, to enhance the explainable quantum architecture search framework [XSF25]. An advanced QAS agent could incorporate quantum processor qubit connectivity and Quantum Information Theoretic metrics directly into its reward function. For instance, an agent could be rewarded not only for achieving high fidelity but also for doing so with a circuit that minimizes a specific entanglement measure or maximizes a desired correlation pattern, keeping in mind the qubit connectivity of the processor. This would enable the agent to learn not just a solution, but a solution with a specific, desirable physical character, thereby transforming QAS from a black-box optimizer into a tool for discovering physically-motivated circuit designs.

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